

Feature: Flexi Cap Mutual Fund Portfolio Analysis Module

Important Implementation Instruction

Do NOT implement this entire feature in one large change.

Implement it in the following 3 phases:

Phase 1: Mutual Fund Data Engine and Validation

Phase 2: Mutual Fund Analysis UI and Analytics

Phase 3: Live Holding Integration

Complete and validate each phase before moving to the next phase.

Do not unnecessarily modify existing working functionality.

BUSINESS OBJECTIVE

My long-term goal is to gradually align part of my stock portfolio with selected Flexi Cap mutual funds.

I do NOT want to blindly replicate mutual fund portfolios or replace all my existing stocks.

I want to build a blended portfolio consisting of:

- Stocks selected using my own investment strategy.
- High-conviction stocks held by Flexi Cap funds.
- Potential value opportunities identified from mutual fund holdings.
- Stocks recently added by fund managers.
- Stocks gradually accumulated by fund managers.
- Stocks consistently held for several months.
- Stocks where multiple Flexi Cap funds show common conviction.

The objective of this module is to create a **decision-support and portfolio-analysis system**.

MUTUAL FUNDS CURRENTLY IN SCOPE

The consolidated source file currently contains monthly portfolio disclosure data for:

1. HDFC Flexi Cap Fund
2. Parag Parikh Flexi Cap Fund
3. Quant Flexi Cap Fund
4. JM Flexicap Fund

The architecture must remain dynamic so additional mutual funds can be added later.

SOURCE FILE LOCATION

The monthly consolidated mutual fund portfolio file will be stored in:

C:\My_Data\Shares Market\Antigravity

Every month I will update the consolidated file by appending the latest month's portfolio disclosure data.

The application must automatically detect newly added months.

Do NOT hardcode months.

Do NOT assume column names before inspecting the actual source file.

PHASE 1: MUTUAL FUND DATA ENGINE AND VALIDATION

Phase 1 Objective

Create a reliable backend/data-processing layer for mutual fund portfolio analysis.

Do NOT start building the full Mutual Fund UI until the data engine has been validated.

STEP 1: INSPECT THE EXISTING PROJECT

Before writing code:

1. Analyse the complete project architecture.
2. Identify the frontend framework.
3. Identify the backend framework.
4. Identify the existing data-loading architecture.
5. Identify how the Live Holding page receives stock data.
6. Identify existing table libraries.
7. Identify existing chart libraries.
8. Identify existing caching mechanisms.
9. Identify existing reusable UI components.
10. Identify existing stock-name normalization or symbol-mapping logic.

Reuse the existing architecture wherever practical.

Do not introduce unnecessary new libraries.

STEP 2: INSPECT THE MUTUAL FUND SOURCE FILE

Inspect the actual consolidated mutual fund file.

Identify:

- File name and file type.
- Sheet names, if Excel.
- Exact column names.
- Fund name column.
- Stock/company name column.
- Month/date column.
- Quantity column.
- LTP/market price column.
- Any ISIN column.
- Any NSE/BSE symbol column.
- Existing market value column, if available.
- Duplicate records.
- Blank values.
- Invalid numeric values.

Before implementing calculations, report the detected schema.

Do not guess column names.

STEP 3: CREATE A MUTUAL FUND DATA MODEL

Create a normalized internal data structure.

The conceptual structure should contain:

- Fund Name
- Fund Code
- Stock Name
- Normalized Stock Name
- ISIN, if available
- Stock Symbol, if available
- Portfolio Month
- Quantity
- LTP
- Holding Value
- Holding Value Crore

Calculate:

$\text{Holding Value} = \text{Quantity} \times \text{LTP}$

$\text{Holding Value Crore} = \text{Holding Value} / 10,000,000$

Use full precision internally.

Round values only for UI display.

All primary mutual fund trend analysis should be based on **Holding Value Crore** unless specifically stated otherwise.

STEP 4: MUTUAL FUND CODES

Create a reusable fund-code mapping.

Current mapping:

HDFC Flexi Cap Fund = H

Parag Parikh Flexi Cap Fund = P

Quant Flexi Cap Fund = Q

JM Flexicap Fund = J

Do not scatter these mappings across multiple files.

Maintain them in one reusable configuration or mapping layer.

The design should allow future funds to be added easily.

STEP 5: STOCK NAME NORMALIZATION

Stock matching is critical.

Create a reusable stock-name normalization function.

At minimum:

- Convert to uppercase.
- Trim spaces.
- Remove duplicate spaces.
- Remove unnecessary punctuation.
- Normalize "&" and "AND".
- Remove common suffixes where appropriate:
 - LIMITED
 - LTD
 - LTD.
- Handle obvious formatting differences.

Matching priority should be:

1. ISIN match, if available.
2. Stock symbol match, if available.
3. Normalized deterministic stock-name match.
4. Explicit manual stock mapping.

Avoid aggressive fuzzy matching.

Do not automatically match stocks when confidence is low.

Create an unmatched-stock report/list.

I want to be able to identify stocks that failed to match.

STEP 6: DATA VALIDATION

Validate the source data for:

- Missing fund name.
- Missing stock name.
- Missing month.
- Blank quantity.
- Blank LTP.
- Invalid quantity.
- Invalid LTP.
- Duplicate fund + month + stock records.
- Unexpected fund names.
- Unexpected month formats.

For duplicate records, inspect the source structure before deciding whether records should be aggregated.

Do not silently remove data-quality issues.

Create clear logging.

STEP 7: CREATE ANALYTICAL DATA FUNCTIONS

Create reusable backend/data-processing functions for:

- Get available mutual funds.
- Get available months.
- Get latest available month.
- Get previous month.
- Get stock holdings by fund.
- Get stock holdings across all funds.
- Calculate 1-month change.
- Calculate 2-month change.
- Calculate 3-month change.
- Calculate percentage change.
- Identify newly added stocks.
- Identify exited stocks.
- Identify continuous increase.
- Identify continuous decrease.
- Identify consistently held stocks.
- Identify common holdings.
- Identify common accumulation.
- Identify divergent fund activity.

Keep business logic separate from frontend components.

CHANGE CALCULATION LOGIC

Month-on-month change:

Current Month Holding Value - Previous Month Holding Value

Percentage change:

$((\text{Current Month Holding Value} - \text{Previous Month Holding Value}) / \text{Previous Month Holding Value}) \times 100$

Handle division by zero safely.

For a newly added stock:

Previous holding = zero or absent.

Mark the stock as:

NEW

Do not display an infinite percentage change.

For an exited stock:

Previous holding > 0

Current holding = zero or absent.

Mark the stock as:

EXITED

TREND DEFINITIONS

Increased in Last 1 Month

Latest month holding > previous month holding.

Increased Continuously for Last 2 Months

Latest > Month -1

AND

Month -1 > Month -2

Increased Continuously for Last 3 Months

Latest > Month -1

AND

Month -1 > Month -2

AND

Month -2 > Month -3

Apply equivalent reverse logic for continuous decreases.

Consistently Held for Last 5 Months

Stock must have a positive holding in every available month within the latest 5-month window.

Also classify the trend as:

- Stable
- Gradually Increasing
- Gradually Decreasing
- Mixed

Define "Stable" using a reasonable configurable tolerance rather than requiring exact equality.

Document the selected tolerance.

PHASE 1 CACHING

The source file will grow every month.

Do not repeatedly reload and recalculate the complete file for every UI interaction.

Implement caching based on source-file modification time or another appropriate project-compatible mechanism.

Reprocess the mutual fund data when the source file changes.

PHASE 1 DELIVERABLE

Before proceeding to Phase 2:

Provide a concise implementation summary containing:

- Source file detected.
- Source schema.
- Number of mutual funds.
- Number of months.
- Earliest month.
- Latest month.
- Number of unique stocks.
- Data-quality issues found.
- Duplicate records found.
- Unmatched stocks, if Live Holding matching was tested.
- Files created or modified.

Validate the core calculations with sample stocks.

Only after Phase 1 is working correctly, proceed to Phase 2.

PHASE 2: MUTUAL FUND ANALYSIS UI AND ANALYTICS

Phase 2 Objective

Create a new main navigation page called:

Mutual Funds

The page must follow the existing project's styling, layout and UI conventions.

PAGE HEADER AND FUND SELECTOR

Add a Mutual Fund dropdown.

Values should dynamically come from the processed data.

Include:

- All Mutual Funds
- HDFC Flexi Cap Fund
- Parag Parikh Flexi Cap Fund
- Quant Flexi Cap Fund
- JM Flexicap Fund

Do not hardcode the fund list in the UI if the backend can provide it dynamically.

KPI SUMMARY CARDS

For an individual mutual fund, show:

- Total Stocks Held.
- Total Portfolio Holding Value.
- New Stocks Added This Month.
- Stocks Increased This Month.
- Stocks Decreased This Month.
- Stocks Exited This Month.

For All Mutual Funds, also show:

- Stocks Held by 2+ Funds.
- Stocks Held by 3+ Funds.
- Stocks Held by All 4 Funds.
- Stocks Accumulated by Multiple Funds.

All KPI calculations should use the latest available month.

MAIN MONTHLY HOLDING TABLE

Create a table with:

Rows = Stock Name

Columns = Available Months

Values = Holding Value in ₹ Crore

Example:

| Stock Name | Jan-26 | Feb-26 | Mar-26 | Apr-26 | 1M Change | 1M % | 2M Change | 2M % | 3M Change | 3M % |

Display months chronologically.

Latest month should appear on the right.

After monthly columns, display:

- 1M Change ₹ Crore
- 1M Change %
- 2M Change ₹ Crore
- 2M Change %
- 3M Change ₹ Crore
- 3M Change %

Use:

- Positive visual indicator for increases.
- Negative visual indicator for decreases.
- Neutral indicator for no material change.
- NEW indicator.
- EXITED indicator.

The table should support:

- Stock search.
- Sorting.
- Column sorting.
- Horizontal scrolling.
- Sticky stock-name column, if practical.
- Tooltips.
- Existing project pagination pattern, if applicable.

ANALYSIS FILTER PANEL

Create an Analysis Filter dropdown or filter panel.

The user should be able to quickly analyse stocks from different perspectives.

Include:

New Stocks

- Added in latest month.
- Added in last 2 months.

- Added in last 3 months.

Decreased Holdings

- Decreased in last 1 month.
- Decreased continuously for last 2 months.
- Decreased continuously for last 3 months.

Increased Holdings

- Increased in last 1 month.
- Increased continuously for last 2 months.
- Increased continuously for last 3 months.

No Change

- Hide stocks with no change in latest month.
- Show only stocks with no change in latest month.

Consistent Holdings

- Consistently held for last 5 months.
- Consistently held and stable.
- Consistently held and gradually increasing.
- Consistently held and gradually decreasing.

High Conviction

- Top 10 holdings.
- Top 20 holdings.
- Top 30 holdings.

Fund Manager Activity

- Biggest Monthly Buying.
- Biggest Monthly Selling.
- Aggressive Accumulation.
- Aggressive Reduction.
- Recently Exited Stocks.

ALL MUTUAL FUNDS ANALYSIS

When All Mutual Funds is selected, enable additional filters.

Common Holdings

- Held by at least 2 funds.

- Held by at least 3 funds.
- Held by all available funds.

Do not hardcode "4 funds" in calculation logic.

Use the total number of active funds dynamically.

Common Accumulation

Identify stocks where at least two mutual funds increased their holding in the latest month.

Show:

- Stock.
- Number of funds accumulating.
- Fund codes.
- Combined increase in ₹ Crore.

Example:

ICICI Bank | 3 Funds | H|P|J | +₹45.5 Cr

Divergent Fund Activity

Identify stocks where:

- At least one mutual fund increased holdings.
- At least one other mutual fund decreased holdings.

Show the fund-level direction.

Example:

HDFC Bank

H ↑

P ↑

Q ↓

J-

ADD A "PORTFOLIO SIGNAL" COLUMN

Add a derived analytical column called:

Portfolio Signal

This is NOT an investment recommendation.

It is only a data-based classification.

Possible values:

- New Entry
- Accumulating
- Strong Accumulation
- Stable Holding
- Reducing
- Strong Reduction
- Exited
- Mixed Fund View

Suggested logic:

Strong Accumulation: Holding increased continuously for 3 months.

Accumulating: Latest month holding increased.

Strong Reduction: Holding decreased continuously for 3 months.

Reducing: Latest month holding decreased.

Stable Holding: Held consistently with movement within the configured stability tolerance.

Mixed Fund View: Applicable to All Mutual Funds when different funds have opposing activity.

Use a tooltip explaining that Portfolio Signal is based only on disclosed mutual fund holding trends and is not a buy/sell recommendation.

STOCK DETAIL VIEW

When I click a stock, open a modal, drawer or detail panel consistent with the existing UI.

Show:

- Stock Name.
 - Selected Mutual Fund.
 - Latest Holding Value.
 - Latest Month Change.
 - Latest Month Change %.
 - First Available Holding Month.
 - Consecutive Months Held.
 - Portfolio Signal.
-

CHART 1: MONTH-WISE HOLDING VALUE

Create a bar chart.

X-axis:

Month

Y-axis:

Holding Value in ₹ Crore

Holding Value = Quantity × LTP

Tooltips should show:

- Month.
 - Quantity.
 - LTP.
 - Holding Value.
-

CHART 2: MONTH-WISE CHANGE IN HOLDING

Create a bar chart.

X-axis:

Month

Y-axis:

Month-on-Month Holding Value Change in ₹ Crore

Calculation:

Current Month Holding Value - Previous Month Holding Value

Clearly distinguish positive and negative values.

Tooltip:

- Month.
- Previous Holding.
- Current Holding.
- Change ₹ Crore.
- Change %.

ALL FUNDS STOCK COMPARISON

When All Mutual Funds is selected and I click a stock, show a fund comparison.

Example:

| Mutual Fund | Latest Holding | 1M Change | 1M % | 3M Trend |

Also show a multi-series month-wise holding chart.

Each series should represent one mutual fund.

Use existing project chart libraries and visual conventions.

PHASE 2 DELIVERABLE

Before moving to Phase 3:

Validate:

- Fund dropdown.
- Dynamic month detection.
- KPI cards.
- Monthly holding table.
- All filters.
- Change calculations.
- NEW logic.
- EXITED logic.
- Common holding logic.

- Common accumulation logic.
- Divergent fund activity.
- Portfolio Signal.
- Stock detail charts.

Provide:

- Files created.
- Files modified.
- APIs/endpoints added.
- Components created.
- Any assumptions made.

Only then proceed to Phase 3.

=====

PHASE 3: LIVE HOLDING INTEGRATION

=====

Phase 3 Objective

Integrate mutual fund portfolio information into the existing:

Live Holding

page.

Do not change existing Live Holding calculations unless required for this feature.

ADD "MUTUAL FUNDS" COLUMN

Add a new column:

Mutual Funds

For every stock in Live Holding, check whether the stock exists in the latest available mutual fund disclosure month.

Use the latest available mutual fund month dynamically.

Do not use the current calendar month unless disclosure data for that month exists.

FUND CODE DISPLAY

Display mutual fund codes:

H = HDFC Flexi Cap Fund

P = Parag Parikh Flexi Cap Fund

Q = Quant Flexi Cap Fund

J = JM Flexicap Fund

Examples:

H

H|Q

P|J

H|P|Q

H|P|Q|J

If the stock is not held by any mutual fund, leave the column blank.

TOOLTIP

When hovering over the fund codes, display full fund names.

Example:

H|P|Q

Tooltip:

H - HDFC Flexi Cap Fund

P - Parag Parikh Flexi Cap Fund

STOCK MATCHING

Use the stock-matching engine created in Phase 1.

Matching priority:

1. ISIN.
2. Stock Symbol.
3. Normalized Stock Name.
4. Manual mapping.

Do not create separate matching logic specifically for Live Holding.

Reuse the centralized matching engine.

OPTIONAL HIGH-VALUE ENHANCEMENT

If it fits naturally within the existing Live Holding table, make the Mutual Funds value clickable.

For example:

H|P|Q

On click, open a compact popup showing:

| Fund | Latest Holding | 1M Change | 3M Trend |

This will allow me to quickly understand mutual fund activity without leaving the Live Holding page.

Implement this only if it does not overcomplicate the existing page.

PHASE 3 VALIDATION

Validate:

- Stocks held by one mutual fund.
- Stocks held by multiple mutual funds.
- Stocks not held by any mutual fund.
- Stock-name formatting differences.
- Latest month detection.

- Tooltip fund names.
- Live Holding page performance.

Create an unmatched Live Holding stock report.

Do not silently fuzzy-match uncertain stocks.

FINAL END-TO-END VALIDATION

After all three phases are completed, perform end-to-end validation.

Test:

1. Append a new month to the consolidated mutual fund source file.
2. Refresh/restart the application as required by the existing architecture.
3. Confirm the new month is automatically detected.
4. Confirm the Mutual Fund table displays the new month.
5. Confirm latest-month calculations move to the new month.
6. Confirm NEW and EXITED stocks are recalculated.
7. Confirm trend filters are recalculated.
8. Confirm common holdings are recalculated.
9. Confirm Portfolio Signals are recalculated.
10. Confirm Live Holding Mutual Funds codes use the new latest month.

Do not hardcode month names, stock names or analytical results.

FINAL IMPLEMENTATION REPORT

After implementation, provide a concise technical report containing:

1. Architecture used.
2. Source file detected.
3. Source schema.
4. Mutual fund data-processing flow.
5. Stock-matching logic.
6. Cache/refresh logic.
7. APIs or backend functions created.
8. Frontend components created.
9. Existing files modified.
10. New files created.
11. Data-quality issues identified.
12. Unmatched stocks.
13. Assumptions made.

14. Known limitations.

The final goal is to create a scalable **Mutual Fund Portfolio Intelligence module** that helps me combine my own stock-selection strategy with insights derived from Flexi Cap mutual fund portfolio activity.

Do not treat the module as an investment recommendation engine.

It is a portfolio research, trend analysis and decision-support tool.